

PROOF OF CONCEPT

SMART GEO-FENCING PROJECT

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ABSTRACT:	The Smart Geo-Fencing project uses GPS and IoT technologies to create a virtual boundary for fishermen. It tracks real-time location data and sends alerts when someone crosses set limits to ensure the safety and security of fishermen.
OBJECTIVE:	To create and test a geo-fencing system that tracks movement within a set area and sends alerts when the boundaries are crossed.
PROPOSED SYTEMS:	The System integrates GPS, Microcontroller (ESP-32), and IOT modules for real-time monitoring and alerts. FLOW DIAGRAM: GPS location → Microcontroller → Geo-fence logic → Alert (buzzer/vibration) → Data sent to cloud/authorities.
PROOF OF CONCEPT:	A prototype was created and tested in a lab. Alerts were successfully generated when someone crossed the geo-fence boundary. For reference; Click Here
CONCLUSION:	This project is validated at TRL 4 (Lab Validation). Future improvements includes AI-based decision-making and deployment for larger-scale smart city and industrial applications.